Online Survey System

Implementation Strategy Summary

Summary focus

- SDLC fit
- Agile methodology
- pilot installation
- ETL migration
- test plan and cases
- validation and review

1. Project Context and SDLC Fit

Current problem

Paper forms, spreadsheets, and email based surveys slow collection, create duplicate records, and weaken reporting.

New system

A centralized Online Survey System supports survey creation, question entry, response submission, result viewing, and export.

Implementation in SDLC

Implementation turns the designed solution into real operational use through installation, migration, testing, and review.

Planning → Analysis → Design → Development → **Implementation** → Review

2. Methodology, Objectives, and Strategy

Chosen methodology: Agile

Short iterations allow the team to refine survey functions after pilot feedback and reduce late stage surprises.

Planning objectives

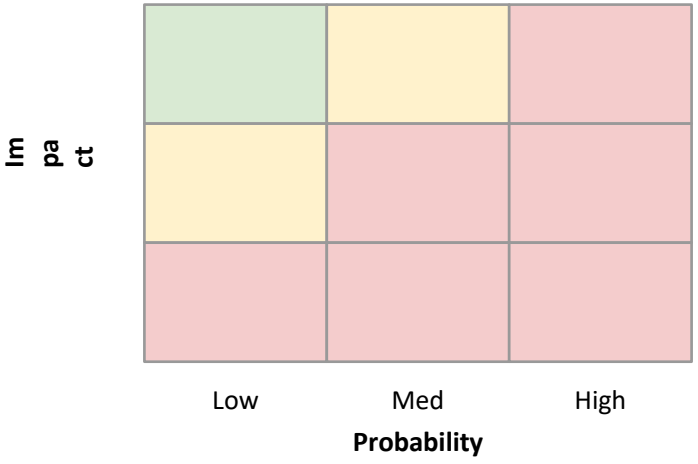
- Improve collection and reporting
- Protect data accuracy during migration
- Keep rollout controlled and low risk
- Verify readiness through structured testing

Implementation strategy

- Confirm scope and ownership
- Prepare environment and data
- Test before pilot release
- Collect pilot feedback
- Review before wider rollout

3. Risk Assessment and Mitigation

Risk matrix



Highest priority risks: changing requirements, migration errors, late defects, and access issues.

R1 Changing requirements

Lock pilot scope and review requirements weekly.

R2 Migration errors

Use backups, trial migration, and count checks.

R4 Late defects

Run unit, integration, system, and acceptance testing.

R3 User resistance

Provide training and collect pilot feedback.

R5 Schedule delay

Track milestones and control scope.

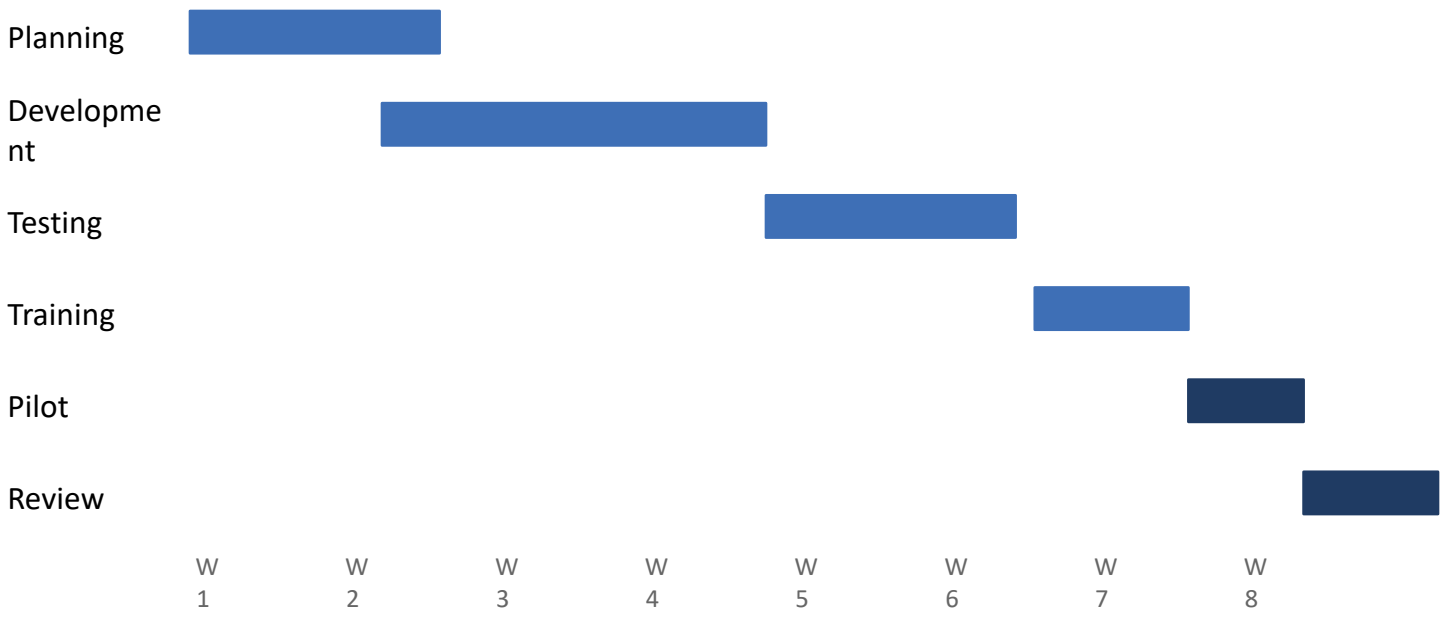
R6 Access issues

Use permissions and password controls.

4. Timeline and Installation

Pilot installation is selected because it reduces risk, allows real user feedback, and limits disruption during the first release.

Implementation timeline



Transition steps

1. Finalize tested build and environment
2. Back up current survey files
3. Clean and load pilot data
4. Train pilot users
5. Launch pilot and capture issues
6. Retest fixes and review next rollout

5. Data Migration Strategy

ETL based migration protects useful historical survey information while preventing poor quality data from entering the new system.

Old Survey Files and Forms → **Extract** → **Transform and Cleanse** → **Load into New System** → Validate Results

Data cleansing focus

Remove duplicates, standardize titles and dates, correct formatting issues, and flag invalid records for review.

Post migration checks

Compare counts, review sample records, confirm export output, and obtain user sign off before approval.

6. Test Plan, Cases, and Tools

Testing levels

Unit → Integration → System → **Acceptance**

Test plan summary

Scope: survey creation, response flow, export, and migration validation.
Objective: prove correct processing and pilot readiness.
Tools: manual testing, JUnit, Selenium, and Postman.
Ownership: developer, tester, and pilot users.

Detailed test cases

TC-1 Login validation: valid users access the menu; invalid users receive a clear error.

TC-2 Survey creation: survey saves correctly and remains editable.

TC-3 Response submission: response is stored and confirmation appears.

TC-4 CSV export: output contains the correct fields and data.

7. Debugging, Data Quality, and Conversion

Debugging and error handling

- Reproduce the issue
- Isolate the affected function
- Fix the defect
- Retest original and related workflows
- Record the outcome

Data quality issues and cleansing

Duplicates: remove repeated records.
Missing values: flag for manual review.
Inconsistent formats: standardize titles, dates, and fields.
Outdated files: migrate only approved relevant records.

Conversion method

Phased conversion aligns with the pilot approach: move approved data and limited use first, validate results, then expand.

8. Validation, Success Factors, and Review

Post migration validation

- compare record counts
- check selected fields and sample records
- confirm migrated surveys open correctly
- verify export outputs
- obtain user confirmation

Key success factors

- clear planning
- clean data
- multi level testing
- active user involvement
- quick correction of defects

Post implementation review

- on time pilot delivery
- critical defects resolved
- migrated data matches approved source records
- users complete main tasks
- manual effort is reduced

9. Lessons Learned and Final Summary

Lessons learned

Implementation success depends on more than coding. It requires planning, migration control, structured testing, user support, and review.

Continuous improvement

Pilot feedback and review findings should guide the next iteration through better workflows, stronger reporting, and clearer training materials.

Implementation Strategy in One View

Plan carefully
Assess risk
Install with a pilot
Migrate through ETL
Test at multiple levels
Validate data
Review and improve